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Author(s): Howard Wainer and Carl M. Francolini
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An Empirical Inquiry Concerning Human Understanding of Two-Variable Color Maps

HOWARD WAINER AND CARL M. FRANCOLINI*

The search for rules for effective graphical display, whether for the purpose of communication, exploration, or reconstitution, has been hampered by the lack of a cohesive body of experimental evidence regarding the parameters of efficacious graphical display. To some extent existing evidence is diverse because of the lack of a coordinating theoretical structure and an allied unified graphical vocabulary. In this article we use the rudiments of Bertin's theory of graphics and his graphical semiology. We also present two experimental studies of Two-Variable Color Maps, which we offer as examples of how questions about the efficacy of a graphical form can be addressed.

KEY WORDS: Two-Variable Color Maps; Psychophysical testing; Statistical graphics.

1. WELTANSCHAUUNG

> Although advice on how and when to draw graphs is available, we have no theory of statistical graphics, nor . . . do we have a systematic body of experimental results to use as a guide.

These remarks in a recent article in *The American Statistician* (Fienberg 1979, p. 165) echo similar statements made more than 50 years ago in the *Journal of the American Statistical Association*. Eells (1926) said that "It is noticeable that all of the [advice] given seem to be purely matters of opinion—none show any evidence of an experimental basis of fact." Eells went on to report an experiment "designed to test by objective means, the validity of the criticisms expressed earlier." Sadly, Eells's work was seriously flawed, and six months later Von Huhn (1927) pointed out several problems that invalidated Eells's conclusions. Others tried to study the same questions (the relative efficacy of bar charts versus pie charts) with a similar lack of success (Croxton and Stryker 1927; Peterson and Schramm 1955; Culbertson and Powers 1959).

In fact, the history of experimental investigation of graphical forms reveals few unqualified successes. The sagacity of Fienberg's remark that "good experiments in this area will tax the minds of the best statisticians, and those well versed in the psychology of perception" (p. 177) is apparent to those who read through this literature in search of aid, or to those who have tried to contribute to it.

Yet we are not pessimistic about being able to draw on existing work to solve certain graphic problems. Perhaps this is because, like Bellow's Charlie Citrine, "I was trained in Chicago to make something of such a scant setting. In Chicago you become a connoisseur of the near-nothing" (Bellow 1973). There are theoretical treatises on graphics available. Bertin (1973, 1977) and Bowman (1968) have developed theoretical structures and language to deal with graphic displays. Both these treatises are admittedly lacking in published evidence, but they do present a place to start. Coupled with these theoretical structures, which provide an orientation, are a number of well-done studies that provide a basis on which to build.

Fienberg (1979) describes the Two-Variable Color Map developed by the Census Bureau and offers some criticisms based on his experience. In this article, we would like to answer Fienberg's cry for more by using the rudiments of Bertin's (1973) theory of graphics and his terminology.[1] We have chosen Bertin's terms in the hope of contributing to the establishment of a standardized graphic vocabulary. For if we do not have a vocabulary with which to discuss graphic concepts, we cannot discuss these concepts in an unambiguous manner. Bertin's words are relatively well defined and uniquely related to graphical concepts.

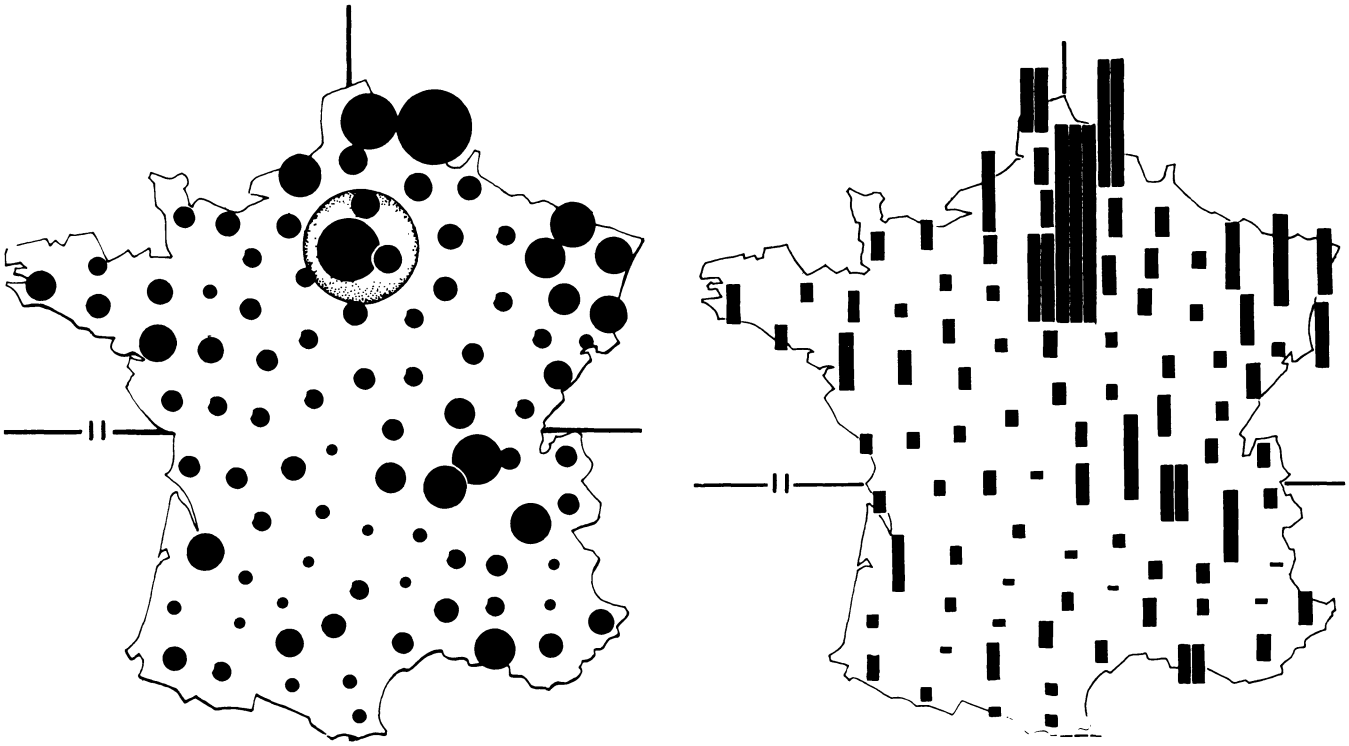
In this article the theoretical structure and graphic vocabulary are oriented around a single problem—the study of Two-Variable Color Maps. In the following sections we shall discuss two experiments that investigate these maps. Although the experiments are relatively simple and straightforward, they become powerful when couched in a theoretical structure. The results of these investigations indicate that careful experimentation is crucial before a new graphical form is adopted over an existing one (Mayr 1874), particularly if it is adopted by a pace-setting organization like Census. A particular form can become conventional simply because of the authority of its originators, and not because of its intrinsic merit.

In performing experiments to test the efficacy of Two-Variable Color Maps we are not denigrating the importance of the opinions of experts. Rather, we completely agree with Macdonald-Ross (1977, p. 371) that "Opinions based upon experience and reason will, in general, prove more reliable than poorly designed experiments. The only weakness of such opinions is the scarcity of public evidence." In fact the raison d'être of this article was well stated by an anonymous referee who remarked that the target of this article

* Howard Wainer is Senior Research Associate at the Bureau of Social Science Research, 1990 M St., N.W., Washington, DC 20036. Carl M. Francolini is a graduate student in the Department of Psychology, Johns Hopkins University, Baltimore, MD 21218. This research was supported by a grant from the National Science Foundation (SOC76-17768) and the U.S. Bureau of the Census. We wish to thank Vincent Barabba, Jacques Bertin, Albert Biderman, Bev Causey, and Bert F. Green, Jr., for their help, support, and suggestions.

[1] Words that have been borrowed from Bertin (1973) and are not obviously identified as such will be demarked by a subscript *b* the first time they appear. These words are taken from a translation of Bertin into English by William J. Berg and Howard Wainer. Any page numbers cited, however, refer to the French text.

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A. and B. Examples of the Use of Ordered Retinal Variables in Displays
(From Bertin 1973, pp. 134, 135. Used by permission.)

richly deserves the criticism it gets, "but opinion and the reflections of individuals on their own experiences . . . have had little evident impact. . . . There is at least hope that experimental evidence will have greater impact."

A detailed description follows of some of the problems that Two-Variable Color Maps were developed to alleviate and two modest experiments that indicate how well they seem to have done it.

2. INTRODUCTION

For many years statisticians and cartographers have been concerned with the effective display of statistical information on a geographic background. To understand the problems, let us consider a map display of a quantitative data component$_b$.[2] The prospective display has three questions to resolve. If the component, say, median family income, is continuous, its graphical representation often requires that it be categorized. How many categories are there? What are the boundaries? How these two questions are answered is crucial to the way the information is depicted. Consider breaking income into the following four categories:

less than $2,000
2,000–2,500
2,501–25,000
more than 25,000

It would be an unusual area of the United States whose income distribution would be depicted in an interesting way with such a categorization. A more usual categorization is

less than $7,000
7,000–9,999
10,000–14,999
15,000 or more

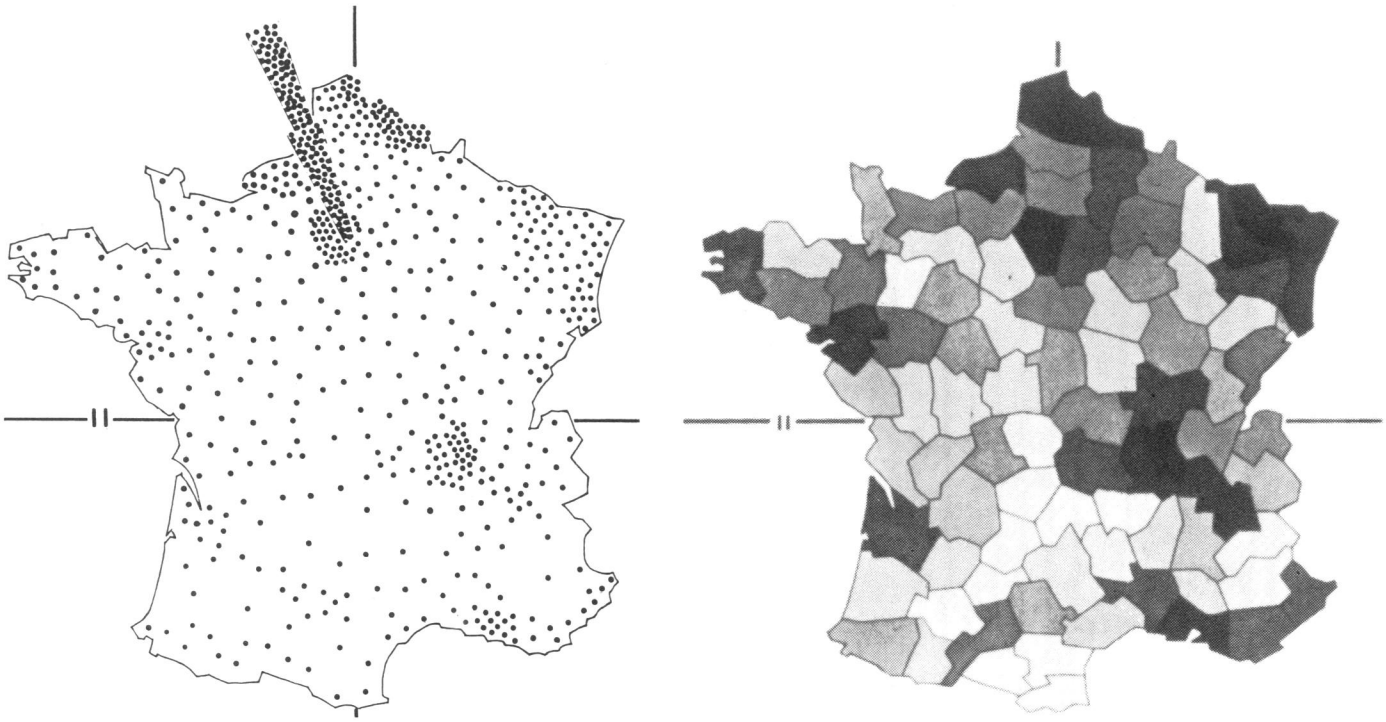
Presumably this classification scheme yields classes that are substantively interesting, and it has sufficient differentiation in the depiction to reflect the distribution of the information accurately. We shall term the particular categories of a quantitative component the *degrees*$_b$ of that component. The number of degrees is the *length*$_b$ of the component. Thus the second categorization has four degrees; it is a component of length four.

The third question to be resolved is how to depict the quantitative component graphically. There have been many schemes proposed that resolve this question

[2] The epistemological basis of this formulation was clearly stated by the Harvard mathematician C.S. Pierce (in Gardner 1978). He felt that all things could be ordered into monads, dyads, and triads. Gardner states,

> Firstness considers a thing all by itself, for example redness. . . . Secondness considers one thing in relation to another, for example a red apple. . . . Thirdness concerns two things "mediated" by a third, for example an apple falling from a tree. The tree and the apple are linked by the relation "falling from". . . . Pierce applied firstness, secondness and thirdness to every branch of philosophy. There is no need, he argued, to go on to fourthness or fifthness and so on, because in almost every case these higher relations can be reduced to combinations of firstness, secondness and thirdness. On the other hand, genuine thirdness can no more be reduced to secondness than can genuine secondness to firstness. (p. 23)

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C. and D. Examples of the Use of Ordered Retinal Variables in Displays
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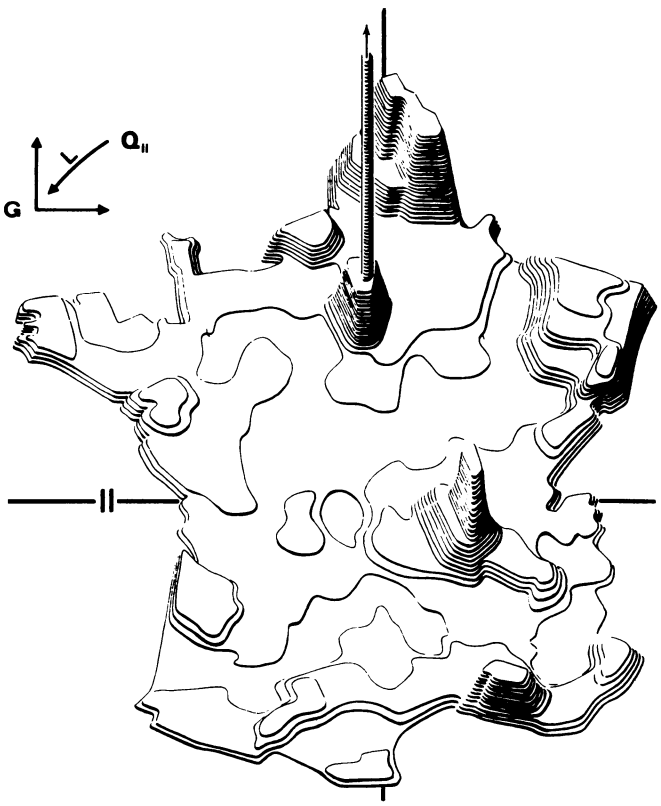
rather well (Williams 1956; Jenks and Knos 1961; Castner and Robinson 1968). Figure A shows dots of various sizes, Figure B bars of various lengths, Figure C dots of various densities, Figure D shading of various darkness; Figure E shows the same effect in a stereogram. All these figures use a natural visual metaphor (what we shall call a *retinal variable*$_b$) to represent the underlying quantitative component. It is a natural metaphor in that the increasing size of the dots represents a natural order that corresponds to the order of the increasing size of the statistical component. Consider how poorly a map in which this correspondence was not maintained would work:

small dot = more than \$15,000
large dot = 10,000–14,999
medium small = 7,000–9,999
medium large = less than \$7,000

That is, the use of a naturally ordered retinal variable in a way that is counter to its order is clearly a mortal error. A more venial sin is to represent an ordered component with a retinal variable that has no obvious natural ordering. In this situation the displayer requires the viewer to memorize a legend and compare his or her perception to that memory. This makes instantaneous comprehension of the data structure difficult and comprehension even after study tedious. An example of such a depiction would be

less than \$7,000 = green
7,000–9,999 = orange
10,000–14,999 = yellow
15,000 or more = blue

One might ask why anyone would use colors to display badly what can be done far better and more cheaply monochromatically. Indeed, there does not seem to be any good reason. Colors do have a natural



E. Example of the Use of Ordered Retinal Variables in Displays
(From Bertin 1973, p. 132. Used by permission.)

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order, the spectrum, but it is not so visually compelling as the monochrome schemes illustrated in Figures A–E. One could, however, use various saturations of color to reflect an ordered data component; this method, very much akin to various shadings in a monochrome scheme, yields much the same visual impact (Cuff 1973). Shown in Figure F is a map in which this approach is tried, with one modification: Cuff's conclusion that only one color should be used is violated by making the lowest degree yellow and representing the next three degrees by three saturations of blue. This scheme seems to work reasonably well, although it does require a modest bit of memorization (yellow = low), which could have been naturally depicted by representing the lowest degree by a maximally unsaturated blue (white). It also seems to set the lowest category perceptually apart from the other three, which may not be justified by the data.

It would seem that a variety of solutions exist for a reasonable display of a single ordered component, what we shall term a *univariate display*. Suppose we have two statistical components that we wish to display simultaneously? This was precisely the question that was addressed by the U.S. Bureau of the Census (1975) in which a Two-Variable Color Map was developed and described. Subsequently, Two-Variable Color Maps were used prominently in Census work, particularly in the short-lived monthly magazine chartbook of social and economic indicators, *STATUS*. The Two-Variable Color Map displays one component as shown in Figure F and a second with varying saturations of red (see Figure G); these two one-component maps are then superimposed and their colors mixed to yield the Two-Variable Color Map shown in Figure H.

Let us now empirically examine the efficacy of this display technique.

3. GOOD FOR WHAT?

Graphical displays can be used to supply answers to questions at three levels of sophistication.[2] These have been termed (Bertin 1973):

1. The Elementary Level—translating the retinal variable back to the quantitative component (e.g., "What is the median family income at this place?");
2. The Intermediate Level—relating trends seen in the retinal variable to some other informational component (e.g., "What is the relationship between median family income and distance from the inner city?");
3. The Superior Level—comparing the entire structure of one component to that of another (e.g., "How does the geographic distribution of median family income relate to that of percentage of high school graduates?").

In this article we shall describe two experiments that investigate how well Two-Variable Color Maps can be read at an elementary level. In particular we shall concern ourselves with how quickly and how accurately one can answer questions like "What is happening at this place?" We shall also investigate the ease with which one can learn the rather complex legend of the Two-Variable Color Map and improve one's performance with practice.

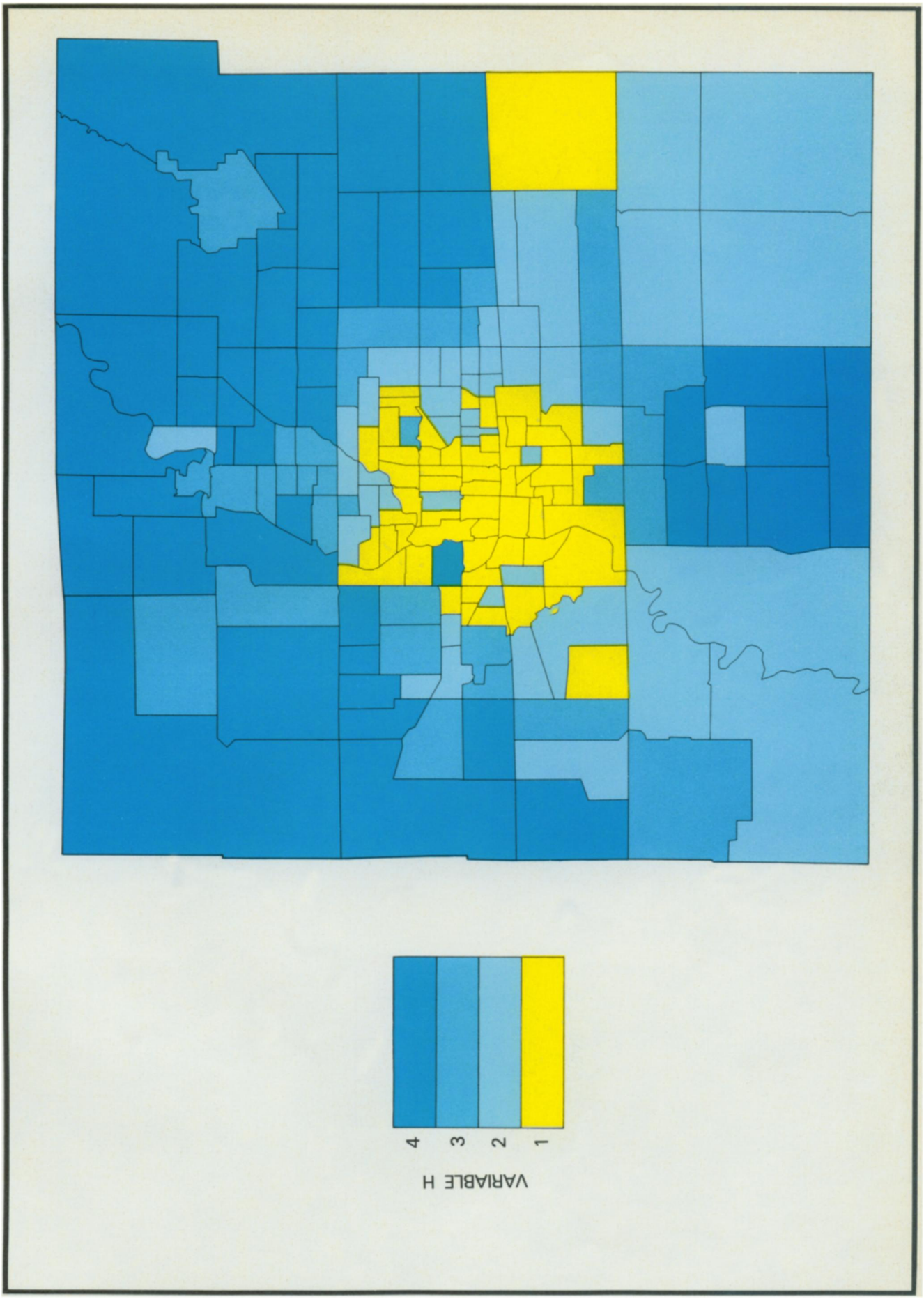
Bertin (1973) in his treatise on graphical methods is clearly pessimistic about the possibility of success of these complex color maps, although he presents no evidence other than his own impressions (and the reader's) for this conclusion. He feels that one cannot make 16 discriminations in color: "Color variation alone yields only about six selective degrees. Beyond that, one must use schemes involving monochrome figurations" (Bertin 1973, p. 325).

The second question is of great importance, for we have repeatedly found that when a variety of graphical forms are compared (Wainer 1974; Wainer and Reiser 1976; Wainer, Groves, and Lono 1978) the biggest effects are (a) individual differences among the subjects and (b) learning. Of course, the displays being compared, although structurally quite different, were competently and unambiguously constructed. In any case, the great effect due to learning tells us that an unfamiliar display may initially fare badly in comparison with a better-known, but in some sense inferior, display, but that after the former is used for a while, its superiority will manifest itself (Wainer and Reiser 1976 and Biderman 1976). For the forms we have tested we have found that performance usually seems to reach an asymptote after only very few exposures. This means that innovative graphics are possible if they only require a modest training period until viewers adapt, and then this innovative form becomes part of the individual's graphical repertoire. If initial performance is poor, however, and the form cannot be learned, its institutionalization should be discouraged.

4. EXPERIMENT 1

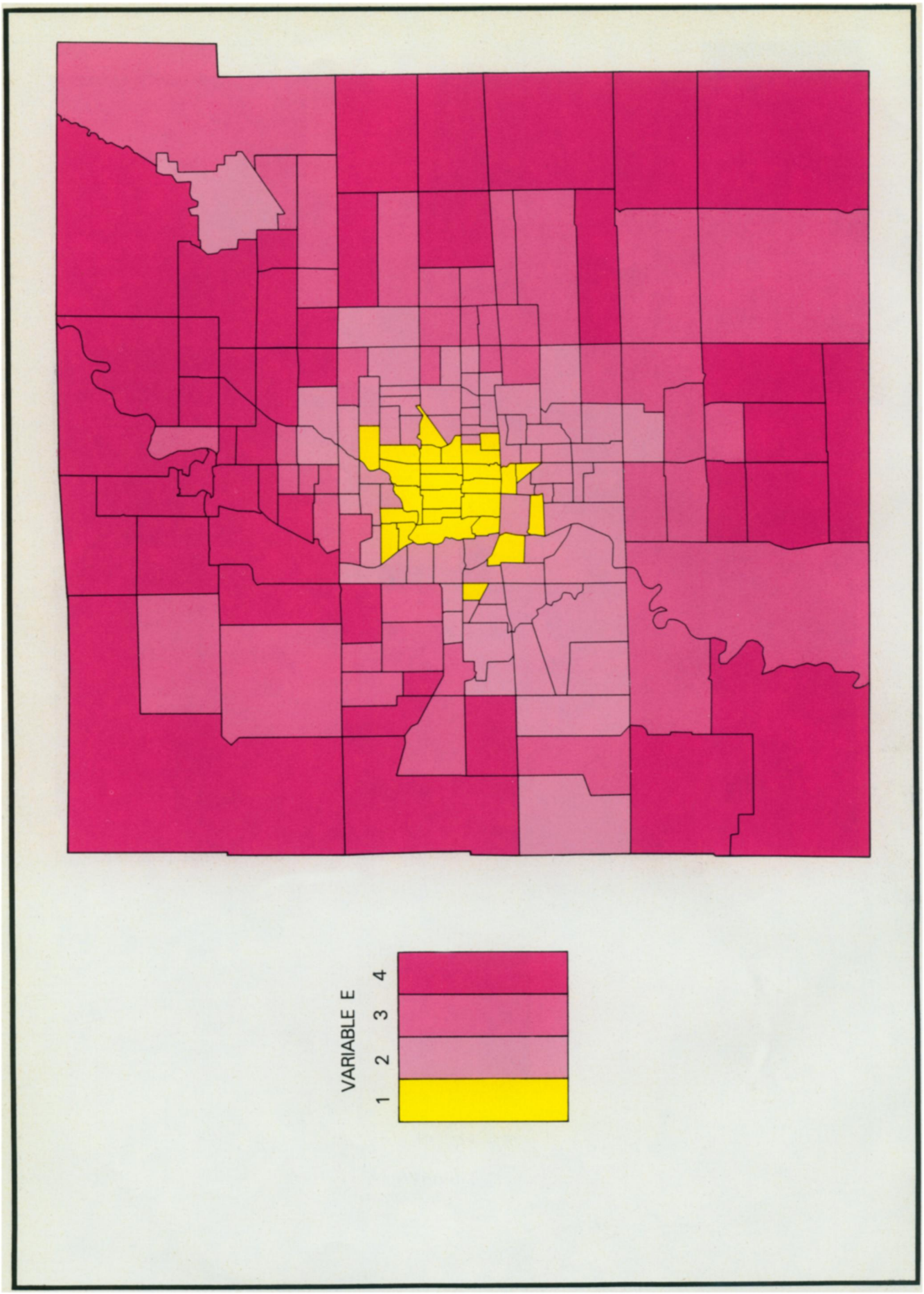
In this experiment we ask subjects, "What is happening at this place?" Two-Variable Color Maps are compared with two corresponding one-variable color maps. We record how long it takes to answer the questions and how many mistakes were made. Of course, had we asked instead, "Where is (x,y) occurring?" (where (x,y) is some bivariate event), the two univariate maps would have been a serious failure. To complete such a task one would have to locate all areas that were, say, yellow on one map and then search for all areas that were, say, dark blue on the other. Then the intersection of these two sets would have to be determined. It is clear that merely looking for bright green on the Two-Variable Maps would be far easier. Thus the superiority of the bivariate maps to the two univariate ones in this task seems so obvious that no further testing is required. If an unacceptably high error rate is observed with a bivariate map in identifying "What is happening at

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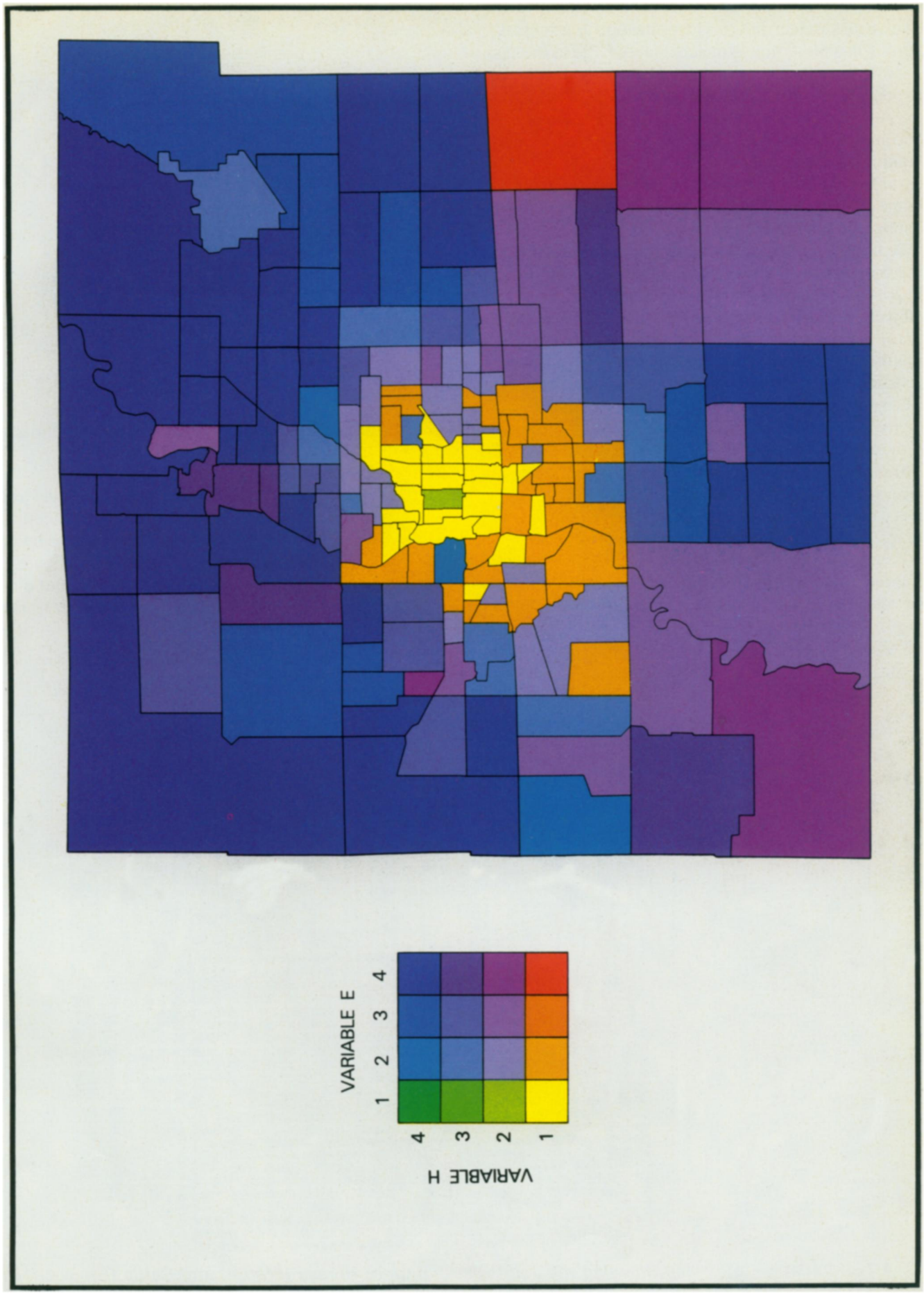
F. Univariate Color Map of Fort Wayne Showing Variable H in Blue (U.S. Bureau of the Census)

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G. Univariate Color Map of Fort Wayne Showing Variable E in Red (U.S. Bureau of the Census)

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H. Two-Variable Color Map Crossing Variables H and E (U.S. Bureau of the Census)

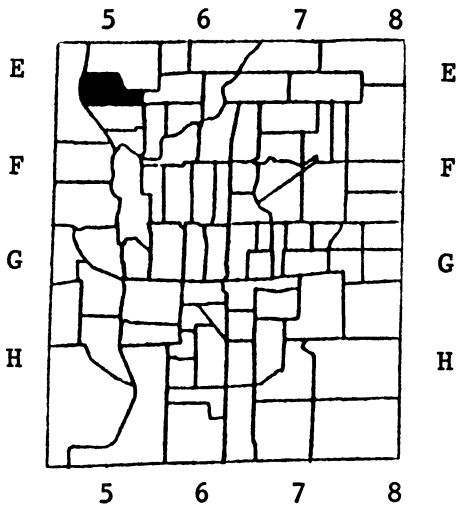
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this place?'' it is not likely to be diminished in the inverse task (which has this identification as a subtask). Thus if this error rate is unacceptably large in the task we tested, an alternative display ought to be developed. To jump ahead for a moment, our results indicate that the performance we expect with these bivariate maps on almost any task is a lot like that expected of a dog walking on its hind legs.

Subjects. Sixteen graduate students in psychology at Johns Hopkins University participated in sessions lasting about 45 minutes.

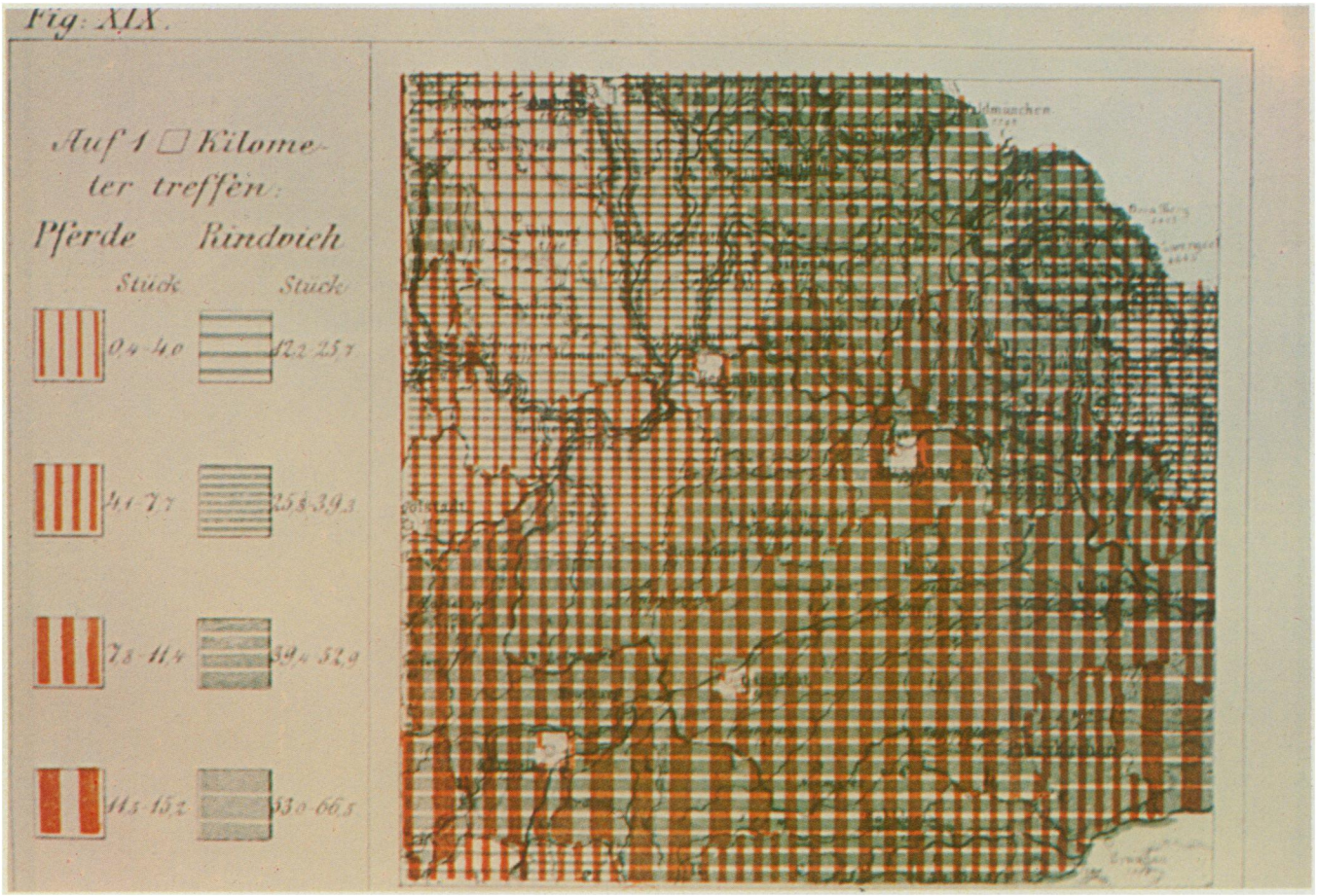
Design. A mixed design with one within factor and one between factor was used. The within factor was map type (bivariate versus univariate) in which all subjects received both types of maps. The between factor pertained to the order in which the subjects received the two types of maps. Subjects were randomly assigned to one of two groups; subjects 1 through 8 received the trials with the bivariate maps first, subjects 9 through 16 received the trials with the univariate maps first.

Stimuli. Examples of the map types are shown here (Figures F, G, and H). These were labeled every half inch, numerically along the y axis and alphabetically along the x axis, yielding a familiar coordinate system for maps. In addition a monochrome rendering of a section of the colored map was also prepared with the specific area in question shaded. Both the letter-number coordinate of the approximate area and the monochrome rendering were presented simultaneously to indicate to the subject exactly what place was being asked about. A sample ''question'' is shown in Figure I.



I. A Sample ''Question'' From the Experiments

All maps were of the same geographic area, but nine different variables were displayed. In addition to the



J. A Two-Variable Color Map Showing the Joint Distribution of Horses (Pferde) and Cattle (Rindvich) in Eastern Bavaria Done According to Scheme 1 (From Mayr 1874.)

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nine distinct univariate maps there were nine combinations of the variables to yield nine different bivariate maps. All maps were supplied by Census and were identical to those used by them.

Procedure. Before the experiment, subjects were read instructions explaining the purpose of the project and were introduced to the two map types. The two map types were divided into two separate booklets containing 9 bivariate maps and 18 (nine pairs) univariate maps, respectively. One pass through an entire booklet is termed a "trial." Each trial was composed of nine "tests." An experimental test consisted of finding the degree of each of the two variables on a specific map type for one particular location. This yielded 18 responses per trial for both map types. Even though a test with the univariate maps required subjects to use two maps, the booklets were arranged so that the same number of page turns were required to pass through the booklets of both map types. Two different variables with only a letter name such as "Variable A" or "Variable B" were used on each test. The univariate booklet contained exactly the same variable pairs that were used with the bivariate map booklet. The color key indicating the degrees of the variables was located on the map sheets directly to the left of the maps. Each different variable pair actually represented different variables, but the respective color keys for both the bivariate and the univariate maps remained the same on each test.

On each trial, subjects were presented with a new cue booklet and an answer sheet containing 18 places for the subject to write in the degrees of each of the two variables for each of the nine tests. Subjects were instructed to flip a page of the cue booklet each time a page of the map booklet was changed.

The location of the specific area to be reported was randomly determined. All subjects received the same location cues. Since some locations were more easy to find because of peculiar shape or location, the same locations and cues were used for both map types. Each subject received four trials of each map type. All four trials with one map type were completed before the four trials with the other map type were begun.

Two dependent variables, response time and number of errors, were recorded. Response time was recorded with a stopwatch from the time the subjects began the first test until the last response was written.

Three practice tests with each map type were presented to subjects before the actual trials. These practice tests enabled the experimenter to determine if the subjects understood the task instructions. The instructions emphasized both speed and accuracy. Because subjects often commented even before the actual experiment that the color discriminations would be more difficult for the bivariate maps, subjects were instructed to make their best guess if confused and to continue working.

Results. The response times for all subjects on all trials are shown in Table 1; the number of errors per trial are shown in Table 2. There are missing response times for four trials; in three cases this was because the timer failed to operate properly and error counts were still obtained. In one case (subject 16, trial 2) the cue booklet and the map booklet became misaligned and neither response times nor error counts were obtained.

A careful examination of the data matrices shown in Tables 1 and 2 indicates quite clearly what has happened. We shall point out the highlights of the analysis without going into too much detail when the analysis was standard. We shall give more information when something that was done deviated from standard practice. We did separate analyses of the response time

1. Experiment 1—Response Times (Seconds/10)

	Map Type											
	Bivariate				Univariate				Bivariate			
	Trial				Trial				Trial			
Subject	1	2	3	4	1	2	3	4	1	2	3	4
1	16	13	13	13	16	12	12	13				
2	15	13	13	13	14	13	12	14				
3	28	20	17	15	18	14	21	13				
4	16	16	13	16	15	16	13	12				
5	19	18	18	16	19	20	19	20				
6	14	14	11	9	15	14	11	10				
7	28	23	20	X	20	22	18	28				
8	15	16	17	11	17	16	19	18				
9					16	12	16	12	14	13	13	10
10					13	12	10	10	11	11	10	8
11					20	20	17	18	14	X	13	12
12					31	22	16	16	21	15	13	13
13					20	13	16	14	16	19	18	18
14					17	15	15	10	X	12	13	11
15					29	24	20	19	22	17	16	16
16					19	19	14	15	18	X	18	16

2. Experiment 1—Errors per Trial (Out of a Possible 18)

	Map Type											
	Bivariate				Univariate				Bivariate			
	Trial				Trial				Trial			
Subject	1	2	3	4	1	2	3	4	1	2	3	4
1	1	1	0	0	0	0	0	0				
2	1	1	2	3	0	0	0	1				
3	0	0	2	0	0	1	0	0				
4	0	3	0	0	0	0	0	0				
5	1	2	2	2	0	0	0	0				
6	5	3	2	11	3	0	0	0				
7	0	2	1	1	0	1	0	0				
8	1	2	4	1	0	3	0	8				
9					0	0	0	0	0	1	3	0
10					0	0	0	0	0	3	2	1
11					0	0	0	0	1	2	4	5
12					0	0	0	0	3	1	0	1
13					2	1	0	2	1	0	0	2
14					0	0	0	0	0	2	2	0
15					3	0	1	0	4	1	1	2
16					2	3	1	3	1	X	0	0

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and the errors. The dependent variables analyzed were the square root of time and arcsin (errors/18)$^{1/2}$, which are reasonably traditional variance equalizing transformations for data of these types. Table 3 shows the F values for the main effects; there were a couple of significant interactions of the time variable ($A \times D$ and $B \times D$), which reflect that how quickly one learns depends on order (which map type was presented first) and also that some people learn faster than others. These interactions are minor, however, and do not enter into our main argument. The principal finding is clear: The biggest differences in response time are individual differences and learning (means for trials 1 through 4 are 13.4, 12.6, 12.3, and 11.8), and the biggest differences in error rates were between map types (bivariate = .24, univariate = .08). Bivariate maps are responded to a bit more quickly but result in much larger error rates. The significant effects due to order probably relate to the number of maps viewed. Subjects who view univariate maps first have seen twice as many maps when they begin work on bivariate maps than have been seen by bivariate-first subjects when they begin work on univariate maps. Also, the subjects who view univariate maps first have the opportunity to practice a more familiar task before going to the less familiar bivariate maps.

3. F Ratios From an Analysis of Variance of Experiment 1

Factor	df	Sin^{-1} (Errors/18)$^{1/2}$	√Time
A. Order	1	.9	15.3
B. Subjects	14	2.9	25.6
C. Map Type	1	34.9	7.8
D. Trials	3	.7	30.6

5. EXPERIMENT 2

The second experiment is virtually identical to the first in procedure. Our aim was to discover to what extent the legend has been internalized by the subjects, since its internalization is clearly necessary for an easy reading of the map at higher levels. Several weeks after the first experiment eight of the same subjects were re-recruited and assigned to one of two experimental groups as before. The experimental procedure was repeated exactly, except that a fifth trial for each map type was added. In this trial the legend was removed, and the subjects had to answer the questions by using the map and their memory of the meaning of the color key. Subjects were told that this would be part of the task before the beginning of the experiment. It should be remembered that by the time the subjects got to this fifth trial they had seen 36 bivariate maps in the first experiment and 36 more in this one. Thus for the bivariate case they had made 72 previous judgments as well as 6 in practice tests. In the univariate case they had seen 72 blue maps and 72 red ones. It was felt that this was a sufficient amount of practice to assure that the subjects had learned as much as they were going to. Subjects 4, 6, 7, and 15 had bivariate maps first; 2, 5, 10, and 13 had univariate maps.

Results. Response times and error counts for each subject and all 10 trials are shown in Tables 4 and 5. The message is clear. There is no significant change in either error rate or response time for the univariate maps when the legend is absent. For the bivariate case response times increase and error rates go through the roof.

Shown in Tables 6 and 7 are some of the results of the analysis. Table 6 clearly indicates that the only difference in error rates is between the two map types, the univariate maps being vastly superior. Once again there are very large individual differences among the subjects in response time, and there are learning effects. There is also an $A \times C$ interaction in the time variable, which reflects the notion that it is a better training procedure to go from the simple to the complex, rather than vice versa. This is the only significant interaction.

Table 7 shows the means of the transformed variables for trials 4 and 5. The increase in response time for trial 5, is significant for bivariate maps. It is not for univariate maps. There is, as suspected from Table 5, a

4. Experiment 2—Response Times (Seconds/10). Fifth Trial Was Without Legend

	Map Type														
	Bivariate					Univariate					Bivariate				
	Trial					Trial					Trial				
Subject	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
4	12	10	10	10	13	12	11	10	9	10					
6	9	9	10	10	14	9	9	12	12	13					
7	13	14	12	12	14	18	15	13	14	16					
15	17	17	12	15	18	21	17	16	16	17					
2						12	19	18	10	13	17	16	13	17	16
5						21	24	20	23	22	21	16	13	14	20
10						10	9	9	7	8	10	9	8	7	12
13						17	16	15	13	16	19	15	14	13	23

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5. Errors per Trial (Out of a Possible 18). Fifth Trial Was Without Legend

	Map Type														
	Bivariate					Univariate					Bivariate				
	Trial					Trial					Trial				
Subject	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
4	0	2	1	2	8	0	0	0	0	0					
6	0	3	3	4	12	1	0	1	0	1					
7	0	1	1	1	1	0	0	0	0	0					
15	2	0	0	2	11	1	0	1	0	1					
2						0	2	0	0	0	0	1	1	0	6
5						0	1	1	0	0	0	2	1	0	5
10						0	3	0	1	0	4	2	2	1	8
15						0	1	1	0	0	0	2	1	0	5

huge increase in error rates from trial 4 to 5 for bivariate maps, none for univariate. An analysis of residuals from the time analysis shows that there is a definite negative association between residuals and number of errors after accounting for factors A, B, C, and D and interactions (if we ignore those who made no errors). The conclusion is that, for those who perform the task less than perfectly, there is a speed/accuracy trade-off. This pattern is clear in Experiment 2, less so in Experiment 1. These results are shown in Table 8.

6. CONCLUSIONS AND DISCUSSION

The results from these two modest experiments point clearly toward one conclusion: The search for an effective display technique for bivariate data on a geographic background must not stop with the current Two-Variable Color Maps. Experiment 1 indicated that reading the Two-Variable Color Map at the elementary level is not possible with acceptable accuracy. Thus, the only purpose for such a map would be for its virtues at the intermediate or superior level. In order for reading at these levels to be possible, however, the legend must be internalized. Experiment 2 showed that memorizing this legend is very difficult, if not impossible. Memorization of the legend is a necessary, though not sufficient, condition for its internalization. To truly internalize the legend and so be able to see its structure more or less spontaneously is more difficult still. Consequently, reading Two-Variable Color Maps at the elementary, intermediate, or superior level is at the very least difficult, and may be impossible.

6. F Ratios From an Analysis of Variance of Experiment 2

Factor	df	$\sin^{-1}(Errors/18)^{1/2}$	$\sqrt{Time}$
A. Order	1	.3	.4
B. Subjects	6	.8	28.7
C. Map Type	1	18.4	1.9
D. Trials	4	.9	3.2

7. Performance Degradation With Removal of the Legend

	Time Means		Error Means	
	Bivariate	Univariate	Bivariate	Univariate
Trial 4	11.0	11.4	.24	.06
Trial 5	12.7	12.0	.67	.06

Note that not being able to read a display at an elementary level does not preclude the possibility of reading it at a higher level. For example, consider a bivariate scatterplot with axes labeled only at their ends "high" and "low." The determination of any particular point cannot be made accurately, but one can quite accurately estimate the degree of relationship between the two displayed components (intermediate level) and compare this structure with that of other bivariate plots (superior level).

The learning of the legend does not mean that the legend is internalized sufficiently for visual comprehension of forms. This is easily demonstrated. Consider a bivariate scatterplot. Suppose we divide it up into small square partitions and instead of showing a dense array of dots within each square we represent the number of dots by their number (in dense areas we plot a 7 or an 8 or a 9, in more sparse areas a 1 or a 2). Of course we must take care to shrink such symbols as 8 so that the amount of ink on the paper is the same as for a 1, and so the perceptual clues from the somewhat graphical character of the Arabic numeral system are removed. There are few symbols that are so well learned as the numbers; yet we contend that such a scatterplot would not offer a very compelling picture of the bivariate structure of the data. This may be one of the reasons that the statisticians in Relles and Rogers's (1977) experiment were only "fairly robust."

Much of our argument about the inutility of Two-Variable Color Maps rests on the notion that the legend scheme for these maps cannot be easily learned, and that even if it could be learned, a spontaneous perception of the structure through such a learned legend is difficult if not impossible. There is some evidence

8. Analysis of Residuals From Time Analysis

No. of Errors	Experiment 1		Experiment 2	
	No. Cases	Residual	No. Cases	Residual
0	68	−.10	37	−.09
1	24	.52	22	.65
2	17	−.50	9	.53
3	11	.65	3	.47
4	3	.37	2	−.08
5	2	.54	1	−2.09
6	0		1	−2.72
7	0		1	−2.52
8	1	−3.73	2	−2.32
9	0		0	
10	0		0	
11	1	−3.06	1	−2.30
12	0		1	−2.55

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that when a coding scheme is very well learned the sort of instantaneous perception useful for the intermediate and superior levels of readings can take place. Chase and Simon (1973) in their study of chess players showed that when a chessboard configuration was shown for a very brief time chess experts could almost perfectly reconstruct the pieces on the board so long as the pieces occurred in positions that could logically have been arrived at during the course of a game. Novices could only reconstruct a few positions in this case, and neither the masters nor the novices were very successful at reconstructing the board when the pieces occurred randomly on the board. Later work (Simon 1978) hints that the explanation for this capacity might lie in the recognition of previously memorized configurations rather than overlearning of the legend.

This chess example is one of a very limited set of experimental evidence we could find that supports, even weakly, the possibility of learning a symbolic legend sufficiently well to allow a spontaneous perception of structure. It must be emphasized that the subjects who manifested such perceptions were beyond the level of casual player and had typically devoted untold hours to the study of chessboards.

It may be that given a similar amount of assiduous study even these Two-Variable Color Maps can yield useful perceptions. If the maps are meant as communicative devices among individuals who are merely acquainted with the display technique and not wedded to it, our earlier conclusion about the necessity for another display scheme still holds.

Our experiments did not test whether someone could see aggregations of same-colored sections. This is another possible use for these maps (e.g., "All these areas go together, they're yellow"). This is a very narrow use, for it can depend crucially on the precise boundaries used to determine the coloring scheme. It would seem that a general recognition of what is identical can be done to some extent, but recognition of which groups are similar cannot ("Is light blue more like yellow than like dark blue?"). Thus Two-Variable Color Maps may have some utility for higher-level reading in spotting groupings of identical areas, but this is so limited that we do not believe it is a feature that should save these maps from being discarded.

Returning now to the question of effective bivariate display of two quantitative components on a geographic background, we shall consider some alternatives that are currently under study.

Scheme 1. Remove yellow as the representative of the lowest degree and substitute the more logically consistent white. Then cross the two colored components.

Scheme 2. Use one of the monochrome figurations described earlier for the representation of one component and vary saturations of color for the second.

Scheme 3. Omit the use of color entirely and use two of the previously described monochrome figurations crossed.

Of course, how well these alternative schemes will work, and under what circumstances, can only be determined empirically. Eliminating one contender may not make the picture clear, but it does make it clearer.

It is interesting to note that Scheme 1 has already been implemented (Mayr 1874) to show the distribution of two kinds of farm animals. Although we have not experimented with this version, it seems clear that this kind of scheme works better than those that violate the rules regarding a natural visual metaphor by having yellow as the lowest degree. It also conforms to rules of accepted wisdom in cartography, many of which have been carefully evaluated empirically.

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